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- 帮我生成word：完整原文（证据句红色加粗 + Q号标注）梳理原文逻辑，在每段后面加上翻译和出题人底层逻辑
- 逐题解析版（完整题干 + 答案红色加粗 + 中文解析 + English Explanation）都不能省略字，一题一答案加详细解析
- 高频词汇 + 同义替换单独列出来，原文不能省略，必须高亮答案句，解析中英文相结合，尽量详细一点。

豆包：

Listening to the Ocean 雅思阅读备考资料

一、完整原文（证据句红色加粗 + Q号标注）+ 段落翻译 + 出题人底层逻辑

Paragraph A

The oceans cover more than 70 per cent of the planet's surface, yet **until quite recently we knew less about their depths than about the surface of the Moon. The Moon has been far more accessible to study because astronomers have long been able to look at its surface, first with the naked eye and then with the telescope, both instruments that focus light (Q1).** Until the twentieth century, however, **no instruments were available for the study of Earth's oceans: light, which can travel trillions of kilometres through the vast vacuum of space, cannot penetrate very far in seawater (Q2).**

段落翻译

海洋覆盖了地球表面70%以上的面积，但直到最近，我们对海洋深处的了解仍少于对月球表面的了解。月球更易于研究，因为天文学家长期以来能观测其表面——最初用肉眼，后来用望远镜，这两种都

是聚光仪器。然而，直到20世纪，人类仍没有研究地球海洋的仪器：光虽能在浩瀚的宇宙真空中传播-数万亿公里，在海水中却无法穿透很远。

出题人底层逻辑

1. 多考点集中段（Q1、Q2，判断题）：

- Q1：通过“far more accessible to study”明确“过去研究月球比海洋容易”，考查“比较关系-理解”；
- Q2：通过“instruments that focus light（月球研究工具）”与“no instruments... light cannot-penetrate（海洋无此类工具）”明确“研究技术不同”，考查“方法差异判断”；

2. 以“月球与海洋研究难度对比”开篇，用“however”强化矛盾，为后文“声音成为海洋研究工具”-铺垫；出题人侧重“比较词（far more accessible, less about... than...）”和“否定词（no instruments）”，这类词汇是判断题的核心信号。

Paragraph B

It turns out that for penetrating water the best instrument is sound. Curious investigators have-long been fascinated by sound and the way it travels in water. As early as 1490, the artist and-scientist Leonardo da Vinci observed: ‘If you cause your ship to stop and place the head of a-long tube in the water and place the outer extremity to your ear, you will hear ships at a great-distance from you.’ **It was not until 1826 that two scientists, Colladon and Sturm, accurately-measured the speed of sound in water. Using a long tube to listen under water (as da Vinci-had suggested), they recorded how fast the sound of a submerged bell travelled across-Lake Geneva in Switzerland (Q9).** What these investigators demonstrated was that water is an-excellent medium for sound, transmitting it almost five times faster than its speed in air.

段落翻译

事实证明，穿透海水的最佳工具是声音。好奇的研究者长期以来对声音及其在水中的传播方式着迷。-早在1490年，艺术家兼科学家列奥纳多·达·芬奇（Leonardo da Vinci）就观察到：“若让船停下，将长管一端放入水中，另一端贴在耳边，你能听到远处船只的声音-。”直到1826年，两位科学家科拉东（Colladon）和斯特姆（Sturm）才准确测量出声音在水中的-传播速度。他们按照达·芬奇的建议，用长管在水下监听，记录了水下铃铛的声音横跨瑞士日内瓦湖-的速度。这些研究者证实，水是声音的优良传播介质，传播速度几乎是空气中的5倍。

出题人底层逻辑

1.

核心考点（Q9，单选题）：通过“Colladon and Sturm, accurately measured the speed of sound in water”明确“研究水中声速的人物”，考查“人物与研究内容的对应”；

2. 以“声音作为海洋研究工具”为核心，通过“达·芬奇的观察→两位科学家的实验”形成时间线，既体现研究的历史延续性，又为Q9设置明确考点；出题人关注“人物（Colladon and Sturm）”和“研究动作（measured the speed of sound）”的绑定，这类“人物-成果”对应是单选题的高频考点。

Paragraph C

A number of factors influence how far sound travels under water and how long it lasts, including particles, salinity, temperature and pressure (Q5). Particles in seawater can reflect, scatter and absorb certain frequencies of sound, just as certain wavelengths of light may be reflected, scattered and absorbed by specific types of particles in the atmosphere (Q7). In 1943, Maurice Ewing and J. L. Worzel conducted an experiment to test the theory that low-frequency waves, which are less vulnerable than higher frequencies to scattering and absorption, should be able to travel great distances, if the sound source is placed correctly. The researchers set off an underwater explosion and learned that it was detected easily by receivers 3,200 kilometres away (Q10). In analysing the results of this test, they discovered a kind of ‘sound pipeline’, known as the ‘deep sound channel’. Sound introduced into this channel of water could travel thousands of kilometres with minimal loss of signal.

段落翻译

多种因素会影响声音在水下的传播距离和持续时间，包括颗粒物、盐度、温度和压力。海水中的颗粒物会反射、散射和吸收特定频率的声音，就像大气中特定类型的颗粒物会反射、散射和吸收特定波长的光一样。1943年，莫里斯·尤因（Maurice Ewing）和J. L. 沃泽尔（J. L. Worzel）开展实验，验证了一项理论：若声源放置得当，低频声波（比高频声波更不易被散射和吸收）应能传播极远的距离。研究者引爆了一次水下爆炸，发现3200公里外的接收器能轻松探测到爆炸声。通过分析实验结果，他们发现了一种“声音管道”，即“深海声道（deep sound channel）”——进入该声道的声音可传播数千公里，信号损失极小。

出题人底层逻辑

1. 多考点集中段（Q5、Q7、Q10）：

- Q5（段落匹配）：通过“including particles, salinity, temperature and pressure”列举“影响声音传播的因素”，考查“列举类信息定位”；
- Q7（段落匹配）：通过“just as”对比“光与声的传播特性相似”，考查“对比类信息定位”；
-

Q10（单选题）：通过“Ewing and Worzel... test the theory... travel great distances”明确“研究声波传播距离的人物”，考查“人物与研究内容的对应”；

2. 以“声音传播的影响因素+实验验证”展开，用“including”明确列举，“just as”强化对比，“conducted an experiment”引出研究；出题人侧重“列举信号（including）”“对比信号（just as）”和“实验主体（Ewing and Worzel）”，这些是段落匹配和单选题的关键考点。

Paragraph D

The US Navy was quick to appreciate the usefulness of low-frequency sound and the deep sound-channel. They developed the Sound Surveillance System (SOSUS), which involved underwater-microphones, called hydrophones, that were placed on the ocean bottom and connected by-cables to on-shore processing centres. **It was Christopher Clark of Cornell University who soon-realised that SOSUS could be used to listen to whales. Using a SOSUS receiver in the West-Indies, he could hear whales that were 1,770 kilometres away** (Q11 铺垫).

段落翻译

美国海军很快意识到低频声音和深海声道的实用价值，研发了“声音监测系统（SOSUS）”——该系统包含被称为“水听器（hydrophones）”的水下麦克风，这些麦克风被放置在海底，通过电缆连接到岸上的处理中心。**康奈尔大学（Cornell University）的克里斯托弗·克拉克（Christopher Clark）很快意识到，SOSUS可用于监听鲸鱼。他在西印度群岛使用SOSUS接收器，能听到1770公里外的鲸鱼声音。**

出题人底层逻辑

1. 隐含考点（Q11 铺垫）：通过“listen to whales 1,770 kilometres away”暗示“SOSUS在鲸鱼研究中的应用”，为Q11（追踪鲸鱼）铺垫；
2. 以“SOSUS系统的研发与初步应用”过渡，连接前文“深海声道理论”与后文“生物、气候研究”，形成“理论→技术→应用”的逻辑链；出题人关注“技术（SOSUS）”与“新应用（listen to whales）”的关联，为后续应用类题目铺垫。

Paragraph E

Whales are the biggest of Earth's creatures, yet these animals are also remarkably elusive. Scientists wishing to observe blue whales must simply wait in their ships for the whales to surface. A few whales have been tracked briefly in the wild in this way but not for very great distances, and much about them remains unknown. But by using SOSUS, **scientists can track the whales-and position them on a map. Moreover, they can track not just one whale at a time, but many-**

creatures simultaneously (Q11). They can also learn to distinguish whale calls; **researchers have detected changes in the calls of finback whales as the seasons change, and have found that blue whales in different regions of the Pacific Ocean have different calls** (Q12).

段落翻译

鲸鱼是地球最大的生物，但也极难追踪。想要观察蓝鲸的科学家，只能在船上等待鲸鱼浮出水面。有人曾用这种方式在野外短暂追踪过少数鲸鱼，但距离不远，关于鲸鱼的很多信息仍未知。但通过SOSUS，**科学家能追踪鲸鱼并在地图上定位，而且不仅能同时追踪一只，还能追踪多只**。他们还能学会区分鲸鱼的叫声：**研究发现，长须鲸（finback whales）的叫声会随季节变化，且太平洋不同区域的蓝鲸叫声也不同**。

出题人底层逻辑

1. 多考点集中段（Q11、Q12，单选题）：

- Q11：通过“track not just one... but many... simultaneously”明确“SOSUS允许同时追踪多只鲸鱼”，考查“细节提取”；
- Q12：通过“changes in the calls of finback whales as the seasons change”明确“长须鲸叫声随季节变化”，考查“细节匹配”；

2. 以“SOSUS在鲸鱼研究中的优势”展开，用“Moreover”强化“多只追踪”的优势，用“as the seasons change”明确叫声变化的原因；出题人关注“复数（many creatures）”和“时间状语（as the seasons change）”，这类细节是单选题的核心考点。

Paragraph F

SOSUS has also proved instrumental in obtaining information crucial to our understanding of climate. The system has enabled researchers to begin making ocean-temperature measurements on a global scale, measurements that are key to understanding the workings of heat transfer between the ocean and the atmosphere. The ocean plays an enormous role in determining air temperature - the heat capacity in only the upper few metres of ocean is thought to be equal to all of the heat in the entire atmosphere (Q6). For sound waves travelling horizontally in the ocean, speed is largely a function of temperature. Thus, the travel time of a wave of sound between two points is a sensitive indicator of the average temperature along its path. **Transmitting sound in numerous directions through the deep sound channel can give scientists measurements spanning vast areas of the globe. Thousands of sound paths in the ocean can be pieced together into a map of global ocean temperatures, and by repeating measurements along the same paths over time, scientists can track changes in temperature over months or years** (Q3, Q13).

段落翻译

SOSUS还被证实能获取对理解气候至关重要的信息。该系统使研究者能在全球范围内测量海洋温度——这些测量对理解海洋与大气间的热交换机制至关重要。海洋在决定气温方面发挥着巨大作用——仅海洋表层几米的热容量，就被认为等同于整个大气的热量总和。对于在海洋中水平传播的声波，其速度很大程度上取决于温度。因此，声波在两点间的传播时间，是衡量沿途平均温度的灵敏指标。通过深海声道向多个方向发射声音，科学家能获得覆盖全球广阔区域的测量数据；将海洋中的数千条声路拼接起来，可绘制出全球海洋温度图；通过长期重复同一声路的测量，科学家能追踪数月或数年内的温度变化。

出题人底层逻辑

1. 多考点集中段（Q3、Q6、Q13）：

- Q3（判断题）：仅说明“声学测温能追踪温度变化”，未与“其他方法耗时对比”，故为NOT GIVEN，考查“无中生有判断”；
- Q6（段落匹配）：通过“ocean-temperature measurements... key to understanding heat-transfer... role in determining air temperature”详述“海洋温度与气候的关联”，考查“因果-类信息定位”；
- Q13（单选题）：通过“track changes in temperature over months or years”明确“SOSUS可-测量温度变化”，考查“细节提取”；

2. 以“SOSUS在气候研究中的应用”展开，用“Thus”建立“声速-温度-气候”的因果链，用“and by repeating...”强化“长期追踪”的优势；出题人关注“因果信号（Thus）”和“否定信息（未提其他方法）”，这些是判断题和单选题的关键考点。

Paragraph G

Researchers are also using other acoustic techniques to monitor climate. Oceanographer Jeff Nystuen, for example, has explored the use of sound to measure rainfall over the ocean. Monitoring changing global rainfall patterns will contribute to understanding major climate-change as well as the weather phenomenon known as El Niño. **Since 1985, Nystuen has used hydrophones to listen to rain over the ocean, acoustically measuring not only the rainfall-rate but also the rainfall type, ranging from drizzle to thunderstorms** (Q4, Q8). By using the sound of rain under water as a ‘natural’ rain gauge, the measurement of rainfall over the oceans will become available to climatologists. In this way, modern society continues to benefit from the investigations of those who, like Leonardo da Vinci, pursued the answers to some basic-questions of nature.

段落翻译

研究者还在使用其他声学技术监测气候。例如，海洋学家杰夫·尼斯图恩（Jeff Nystuen）探索了用声音测量海洋上空降雨量的方法。监测全球降雨模式的变化，有助于理解重大气候变化及“厄尔尼诺（El Niño）”天气现象。自1985年以来，尼斯图恩一直用水听器监听海洋上空的降雨，通过声学方法不仅能测量降雨率，还能区分降雨类型——从细雨到雷阵雨。将水下雨声作为“天然”雨量计，气候学家就能获取海洋上空的降雨量数据。就这样，现代社会仍在受益于像达·芬奇这样的研究者——他们致力于探索自然的基本问题。

出题人底层逻辑

1. 多考点集中段（Q4、Q8）：

- Q4（判断题）：通过“measuring... the rainfall type, ranging from drizzle to thunderstorms”明确“水听器能区分不同降雨类型”，考查“细节匹配”；
- Q8（段落匹配）：通过“Since 1985”“rainfall type, ranging from drizzle to thunderstorms”明确“长期研究不同天气类型”，考查“时间+分类类信息定位”；

2. 以“其他声学技术（测降雨）”收尾，用“for example”举例，“Since 1985”强调长期，“-ranging from... to...”列举降雨类型；出题人关注“时间信号（Since 1985）”和“分类信号（-ranging from... to...）”，这些是判断题和段落匹配题的关键考点。

二、逐题解析版（完整题干 + 答案红色加粗 + 中文解析 + English Explanation）

Questions 1–4（判断题）

题干共性说明

Do the following statements agree with the information given in Reading Passage 1? In boxes 1–4 on your answer sheet, write TRUE if the statement agrees with the information FALSE if the statement contradicts the information NOT GIVEN if there is no information on this

1. In the past, it was easier for scientists to study the Moon than the oceans.

- **完整题干：**过去，科学家研究月球比研究海洋更容易。
- **答案：**TRUE
- **中文解析：**根据Paragraph A，“直到最近，我们对海洋深处的了解仍少于对月球表面的了解。月球更易于研究（The Moon has been far more accessible to study）”。“far more accessible” 直接对应 “easier”，“until quite recently” 对应 “in the past”，与题干表述一致，故为TRUE。
- **English Explanation：** According to Paragraph A, "The Moon has been far more accessible to study until quite recently, and we knew less about ocean depths than the Moon's surface". "Far more accessible" matches "easier", and "until quite recently" matches "in the past", so the answer is TRUE.

2. Techniques for investigating the Moon are the same as techniques for researching the ocean.

- **完整题干：**研究月球的技术与研究海洋的技术相同。
- **答案：**FALSE
- **中文解析：**根据Paragraph A和B，研究月球的技术是“聚光仪器（肉眼、望远镜，instruments that focus light）”，而研究海洋的技术是“声音（sound）”——“光在海水中无法穿透很远（light cannot penetrate very far in seawater），声音是最佳工具（the best instrument is sound）”。- 两者技术完全不同，与题干“the same”矛盾，故为FALSE。
- **English Explanation：** According to Paragraphs A and B, Moon research uses "light-focusing-instruments (naked eye, telescope)", while ocean research uses "sound" (light cannot penetrate seawater). The techniques are completely different, contradicting "the same", so the answer is FALSE.

3. Measuring temperature changes in the ocean using sound is more time-consuming than other methods.

- **完整题干：**用声音测量海洋温度变化比其他方法更耗时。
- **答案：**NOT GIVEN
- **中文解析：**根据Paragraph F，仅说明“用声音可测量海洋温度（sound waves... indicator of temperature）”“长期追踪温度变化（track changes over months or years）”，但未提及“其他方法（other methods）”的耗时，也未将两者对比。题干信息无依据，故为NOT GIVEN。
- **English Explanation：** According to Paragraph F, the text only states "sound can measure ocean temperature and track changes over time" but provides no information about "other methods" or their time consumption. There is no basis for the comparison, so the answer is NOT GIVEN.

4. Hydrophones can distinguish different kinds of rain.

- **完整题干：**水听器能区分不同类型的降雨。
- **答案：**TRUE
- **中文解析：**根据Paragraph G，“尼斯图恩用水听器监听降雨，通过声学方法不仅能测量降雨率，还能区分降雨类型——从细雨到雷阵雨（acoustically measuring... the rainfall type, ranging from drizzle to thunderstorms）”。 “distinguish different kinds of rain” 对应 “measuring the rainfall type, ranging from drizzle to thunderstorms”，与题干表述一致，故为TRUE。
- **English Explanation：** According to Paragraph G, "Nystuen used hydrophones to measure rainfall type, ranging from drizzle to thunderstorms". "Distinguish different kinds of rain" matches "measure the rainfall type", so the answer is TRUE.

Questions 5–8（段落匹配）

题干共性说明

Reading Passage 1 has seven paragraphs, A–G. Which paragraph contains the following information? Write the correct letter, A–G, in boxes 5–8 on your answer sheet. NB You may use any letter more than once.

5. examples of things that affect the distance sound can travel in water

- **完整题干：**影响声音在水中传播距离的事物例子
- **答案：**C
- **中文解析：**根据Paragraph C，“多种因素影响声音在水下的传播距离，包括颗粒物、盐度、温度和压力（A number of factors influence how far sound travels... including particles, salinity, temperature and pressure）”。“including”后列举的“particles, salinity, temperature and pressure”是“影响传播距离的事物例子”，与题干对应，故对应段落C。
- **English Explanation：** According to Paragraph C, "factors influencing sound's travel distance in water include particles, salinity, temperature and pressure". These are "examples of things that affect the distance sound can travel", so the answer is C.

6. details of the connection between ocean temperatures and climate

- **完整题干：**海洋温度与气候关联的细节
- **答案：** F
- **中文解析：**根据Paragraph F，详细阐述两者关联：①海洋温度测量对理解“海洋与大气热交换（heat transfer between ocean and atmosphere）”至关重要；②海洋热容量影响气温（ocean plays an enormous role in determining air temperature）；③用声音测温度可追踪气候相关的温度变化（track changes in temperature for climate study）。这些细节直接对应题干，故对应段落F。
- **English Explanation：** According to Paragraph F, it details the connection: ① ocean temperature measurements help understand ocean-atmosphere heat transfer; ② ocean heat capacity affects air temperature; ③ sound-based temperature measurement tracks climate-related changes. These details match the question, so the answer is F.

7. details of ways in which light and sound are similar

- **完整题干：**光与声相似之处的细节
- **答案：** C
- **中文解析：**根据Paragraph C，“海水中的颗粒物会反射、散射和吸收特定频率的声音，就像大气中特定颗粒物会反射、散射和吸收特定波长的光一样（particles in seawater reflect/scatter/absorb sound, just as particles in atmosphere reflect/scatter/absorb light）”。“just as”明确两者在“被颗粒物反射、散射、吸收”上的相似性，与题干对应，故对应段落C。
- **English Explanation：** According to Paragraph C, "particles reflect, scatter and absorb sound in seawater, just as they do to light in the atmosphere". "Just as" clarifies the similarity, which matches the question, so the answer is C.

8. a reference to a long-term study of different types of weather

- **完整题干：**提及对不同天气类型的长期研究
- **答案：** G
- **中文解析：**根据Paragraph G，“自1985年以来（Since 1985），尼斯图恩用水听器监听降雨，区分从细雨到雷阵雨的不同类型（rainfall type, ranging from drizzle to thunderstorms）”。“Since-1985”体现“长期（long-term）”，“drizzle to thunderstorms”体现“不同天气类型（different types of weather）”，与题干对应，故对应段落G。
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English Explanation: According to Paragraph G, "since 1985 (long-term), Nystuen studied-rainfall types ranging from drizzle to thunderstorms (different weather types)". This matches the-question, so the answer is G.

Questions 9–13 (单选题)

9. According to the passage, who conducted research into the rate-at which sound travels in water?

A Leonardo da Vinci B Colladon and Sturm C Ewing and Worzel D Jeff Nystuen

- **完整题干：**根据文章，谁开展了关于声音在水中传播速度的研究？
- **答案：** B
- **中文解析：**根据Paragraph B，“1826年，科拉东和斯特姆准确测量出声音在水中的传播速度（-Colladon and Sturm, accurately measured the speed of sound-in water）”。 “rate at which sound travels” 对应 “speed of sound-in water”，故答案为B。错误选项排除：A（达·芬奇仅观察到水下听声音，未测速度）；C（尤因-和沃泽尔研究声波传播距离，非速度）；D（尼斯图恩研究降雨，非声速）。
- **English Explanation:** According to Paragraph B, "Colladon and Sturm accurately measured the speed of sound in water in 1826". "Rate at which sound travels" matches "speed of sound", so-the answer is B. Elimination: A (da Vinci only observed underwater sound, not speed); C (Ewing-and Worzel studied sound' s travel distance, not speed); D (Nystuen studied rainfall, not sound-speed).

10. According to the passage, who conducted research into the-distances certain types of sound waves travel in water?

A Leonardo da Vinci B Colladon and Sturm C Ewing and Worzel D Christopher Clark

- **完整题干：**根据文章，谁开展了关于特定类型声波在水中传播距离的研究？
- **答案：** C
- **中文解析：**根据Paragraph C，“1943年，尤因和沃泽尔开展实验，验证低频声波能传播极远的距离-（Ewing and Worzel... test the theory that low-frequency waves... travel great distances）-”。“distances certain types of sound waves travel” 对应 “low-frequency waves... travel-great distances”，故答案为C。错误选项排除：A（达·芬奇未研究传播距离）；B（科拉东和斯特姆研究声速，非距离）；D（克拉克用SOSUS听鲸鱼，非研究声波传播距离）。
-

English Explanation: According to Paragraph C, "Ewing and Worzel tested that low-frequency-waves can travel great distances in water". "Distances certain sound waves travel" matches their-research, so the answer is C. Elimination: A (da Vinci didn't study distance); B (Colladon and-Sturm studied speed, not distance); D (Clark listened to whales, not sound distance).

11. SOSUS allows whale researchers to

A follow a number of whales at the same time. B protect whales as they migrate. C imitate whale-calls of different species. D change the whales' direction of travel.

- **完整题干:** SOSUS允许鲸鱼研究者做什么?
- **答案:** A
- **中文解析:** 根据Paragraph E, “通过SOSUS, 科学家不仅能同时追踪一只鲸鱼, 还能追踪多只 (track not just one whale at a time, but many creatures simultaneously)”。 “follow a number of-whales at the same time” 对应 “track many creatures simultaneously”, 故答案为A。错误选项-排除: B (保护迁徙鲸鱼)、C (模仿鲸鱼叫声)、D (改变鲸鱼行进方向) 原文均未提及, 无依据。
- **English Explanation:** According to Paragraph E, "SOSUS allows scientists to track many whales-simultaneously". "Follow a number of whales at the same time" matches "track many creatures-simultaneously", so the answer is A. Elimination: B (protect migrating whales), C (imitate whale-calls), D (change whales' direction) are not mentioned.

12. Finback whale calls change

A when scientists track them. B at different times of year. C when whales communicate with other-species. D when whales come to the surface.

- **完整题干:** 长须鲸的叫声在什么情况下会变化?
- **答案:** B
- **中文解析:** 根据Paragraph E, “研究者发现, 长须鲸的叫声会随季节变化 (changes in the calls-of finback whales as the seasons change)”。 “at different times of year” 对应 “as the-seasons change”, 故答案为B。错误选项排除: A (被科学家追踪时)、C (与其他物种交流时)、-D (浮出水面时) 原文均未提及, 无依据。
- **English Explanation:** According to Paragraph E, "finback whale calls change as the seasons-change". "At different times of year" matches "as the seasons change", so the answer is B.- Elimination: A (when tracked), C (when communicating with other species), D (when surfacing-) are not mentioned.

13. SOSUS allows scientists to

A make accurate maps of the ocean floor.B measure water-level changes.C investigate ocean-currents.D measure variations in temperature.

- **完整题干：**SOSUS允许科学家做什么？
- **答案：**D
- **中文解析：**根据Paragraph F，“通过SOSUS，科学家能绘制全球海洋温度图，追踪数月或数年内的-温度变化（track changes in temperature over months or years）”。“measure variations in-temperature”对应“track changes in temperature”，故答案为D。错误选项排除：A（绘制海底-地图）、B（测量水位变化）、C（研究洋流）原文均未提及，无依据。
- **English Explanation：**According to Paragraph F, "SOSUS allows scientists to track changes in-ocean temperature over time". "Measure variations in temperature" matches "track changes in-temperature", so the answer is D.Elimination: A (map the ocean floor), B (measure water-level-changes), C (investigate currents) are not mentioned.

三、高频词汇 + 同义替换

（一）高频词汇（词性 + 中文释义 + 原文出处 + 高亮答案句）

词汇（Word）	词性（Part of-Speech）	中文释义（-Chinese Definition）	原文出处（-Source Paragraph）	高亮答案句（-Highlighted Answer Sentence）
accessible	adj.	易获取的；易研-究的	A	The Moon has been far more accessible to study because astronomers have long been able to look at its surface.
salinity	n.	盐度（海水含盐-量）	C	A number of factors influence how

				far sound travels... including particles, salinity , temperature and pressure.
hydrophone	n.	水听器（水下麦克风）	D/G	They developed the Sound Surveillance System (SOSUS), which involved underwater microphones, called hydrophones ..- . / Since 1985, Nystuen has used hydrophones to listen to rain over the ocean.
finback whale	n.	长须鲸（一种大型鲸鱼）	E	researchers have detected changes in the calls of finback whales as the seasons change.
heat capacity	n.	热容量（物质储存热量的能力）	F	the heat-capacity in- only the upper- few metres- of ocean is- thought to be- equal to all- of the heat-

				in the entire-atmosphere.
drizzle	n.	细雨；毛毛雨	G	acoustically measuring not only the rainfall rate but also the rainfall type, ranging from drizzle to thunderstorms.
acoustic	adj.	声学的；声音的	G	Researchers are also using other acoustic techniques to monitor climate.

(二) 同义替换 (原文词 + 替换词 + 对应题目/出处 + 高亮对应句)

原文词 (Original Word)	同义替换 (-Synonym)	对应题目/出处 (-Relevant Question/Source)	高亮对应句 (-Highlighted Matching Sentence)
far more accessible-to study (更易于研究)	easier to study (更容易研究)	Q1 (A)	The Moon has-been far more-accessible to-study – this-matches "easier to-study".
instruments that-focus light vs sound-(聚光仪器 vs 声音)	techniques are the-same (技术相同)	Q2 (A/B)	no instruments-were available for-the oceans... the-best instrument-is sound -- this contradicts "-

			techniques are the- same".
measuring rainfall type, ranging from drizzle to thunderstorms (区- 分细雨到雷阵雨的降- 雨类型)	distinguish different kinds of rain (区分不同类型- 的降雨)	Q4 (G)	acoustically measuring... the rainfall type, ranging from drizzle to thunderstorms – this matches "distinguish different kinds of rain".
including particles, salinity, temperature and pressure (包括颗粒- 物、盐度、温度、压- 力)	examples of things that affect sound’ s travel distance (影响声音- 传播距离的事物例子-)	Q5 (C)	A number- of factors- influence how- far sound travels- ... including- particles, salinity- , temperature and- pressure – these- are "examples of- things that affect- sound’ s travel- distance".
ocean-temperature measurements... key to heat transfer (海洋温度- 测量对热交换关键)	connection between ocean temperatures and climate (海洋温度- 与气候的关联)	Q6 (F)	SOSUS... making ocean-- temperature measurements... key to understanding heat transfer between ocean and atmosphere – this is the "connection between ocean temperatures and climate".

particles reflect/scatter/- absorb sound just as they do to light (颗粒物反射/- 散射/吸收声与光)	ways in which- light and sound- are similar (光与声- 的相似之处)	Q7 (C)	Particles in- seawater can- reflect, scatter- and absorb sound- , just as certain- wavelengths of- light may- be reflected,- scattered and- absorbed – this- shows "similarities- between light and- sound".
since 1985... rainfall type from drizzle to thunderstorms (19- 85年以来...细雨到雷- 阵雨)	long-term study- of different types- of weather (不同天- 气类型的长期研究)	Q8 (G)	Since 1985,- Nystuen has used- hydrophones to- listen to rain...- measuring rainfall- type from drizzle- to thunderstorms- – this is a "- long-term study of- different weather- types".
measured the- speed of sound- in water (测量水中- 声速)	research into the- rate at which sound- travels in water (研- 究水中声速)	Q9 (B)	Colladon and- Sturm, accurately- measured the- speed of sound- in water – this- matches "research- into the rate- at which sound- travels".
tested low-frequency waves travel great distances (验证低频- 声波传播远距)	research into- the distances- certain sound- waves travel (研究- 特定声波传播距离)	Q10 (C)	Ewing and Worzel- ... test the- theory that low- -frequency waves- ... travel great- distances – this-

			matches "research-into the distances-certain sound-waves travel".
track many creatures simultaneously (同时追踪多只生物)	follow a number-of whales at the-same time (同时追踪多只鲸鱼)	Q11 (E)	they can track not-just one whale-at a time, but-many creatures-simultaneously -- this matches "- follow a number of-whales at the same-time".

说明

- 1. 上述内容可直接复制至Word文档，通过“字体颜色”功能将标注为红色加粗的证据句/答案句设置为红色加粗，确保考点清晰；
- 2. 高频词汇与同义替换均关联题目，高亮句直接来自原文，帮助考生精准定位考点，强化“题干-原文” - 的对应思维，适配雅思阅读细节题备考需求。

(豆包AI生成)